Software Engineering - Practical / Oral Exam

Process Model Questions: Q8, Q13, Q14, Q15 (Answers + how to write them in the answer sheet)

Quick summary

Q	Case study	Model	Clue in the question
8	Online Examination System	**Prototyping**	Requirements may change based on feedback
13	Hospital Management System	**Waterfall**	Clearly defined requirements, major changes not expected
14	Mobile Banking Application	**Spiral**	High security, extensive testing, risk management
15	AI-Based Smart Traffic System	**Agile (Scrum)**	AI + IoT, evolving requirements

The 7-step format to follow for every question

- **Write the question number and title** at the top, then the case study in one line.
- **Underline keywords** from the case study (for example: 'requirements change', 'risk', 'clearly defined'). The model is always hidden in these words.
- **State the model clearly** in one line, so the examiner finds it quickly.
- **Justify in 3-4 points** and use words from the case study in every point. This carries the most marks.
- **Explain the phases** briefly, linked to the project (not copied from the textbook).
- **Draw the diagram** with a pencil and ruler, label every box, and show arrows and loops. Write the figure name below it.
- **Add 2-3 advantages and disadvantages** and finish with a one-line conclusion.

Rule of thumb for the keyword to model match

If the question says...	Write this model
Fixed / clear requirements, no change expected	Waterfall
Requirements unclear or change after user feedback, UI-heavy	Prototyping
High risk, security, large and costly project	Spiral
Evolving requirements, frequent delivery, customer involvement	Agile (Scrum)
Core product first, features added in releases	Incremental

Exam tips

- Spend about 12-15 minutes per question: 3 min to read and decide, 5 min on the diagram, 5 min on writing.
- If your faculty taught slightly different diagram styles, follow the teacher's style. The structure above stays the same.
- Do not copy long definitions. Marks go to the case study link, a correct diagram and clear labels.
- For Q15, if your class notes treat Incremental or Iterative as the answer, you may mention it as a related option, but write Agile as the main one.

Note: These are model answers prepared for study. Rewrite them in your own words in the exam.

Q 8. Online Examination System

Question: A college plans to develop an Online Examination System to conduct online tests, evaluate answers, generate results, and maintain student records. The requirements may change based on feedback from faculty and students. Identify a suitable software process model and draw its diagram.

Step 1 - Underline these keywords in the case study

online tests, evaluate answers, generate results, student records, requirements may change, feedback from faculty and students.

Step 2 - Write the selected model

Selected Software Process Model: Prototyping Model

Step 3 - Justification (write 3-4 points)

- The requirements are not fixed and will change after feedback, so a working sample must be shown to users early.
- Faculty and students can see the login page, test screen and result view in the prototype and say exactly what they want.
- Misunderstandings about requirements are found early, so the risk of building the wrong system is lower.
- An online exam system depends heavily on its user interface, which is easy to demonstrate through a prototype.

Step 4 - Phases / working of the model

Phase	Activity (with respect to this case study)
1. Requirements gathering	Collect basic needs: question types, timer, auto-evaluation, result reports.
2. Quick design	Rough design of input and output screens (login, test page, result page).
3. Build prototype	Make a working sample with limited features (sample test and sample result).
4. User evaluation	Faculty and students use it and give feedback.
5. Refine	Update requirements and prototype. Repeat steps 1-4 until accepted.
6. Final system	Design, implement, test, deploy and maintain the full system.

Step 5 - Draw the diagram (neat, labelled, with arrows)

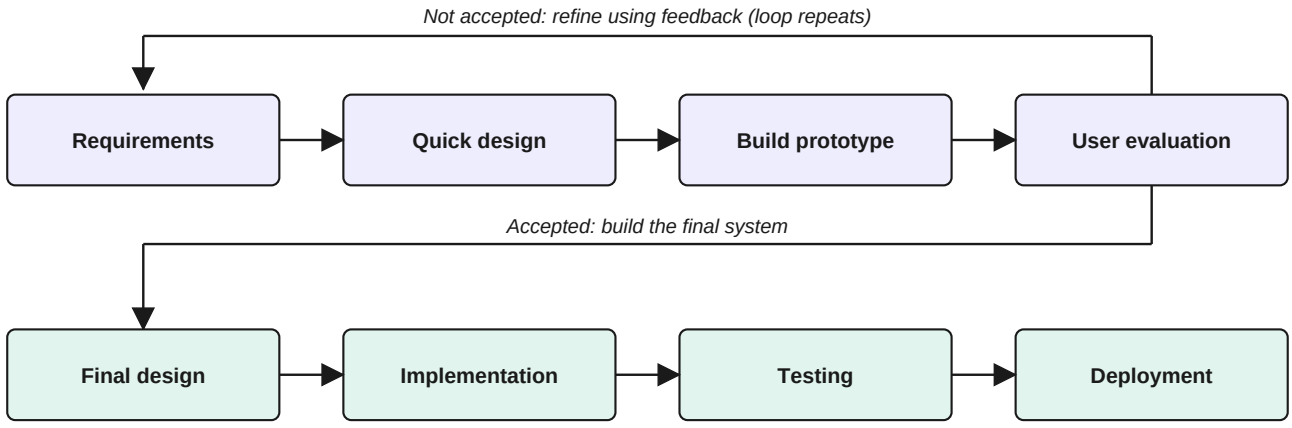

Fig: Prototyping Model

Step 6 - Advantages and disadvantages (2-3 each)

Advantages	Disadvantages
- Early user feedback - Fewer requirement errors - Better user satisfaction	- Too many changes can raise time and cost - Documentation is often weak - Users may mistake the prototype for the final product

Step 7 - Conclusion (last line of the answer)

Prototyping Model is the most suitable model for the Online Examination System, because the requirements change with faculty and student feedback and users can approve the system early.

Viva ready: why not other models?

Waterfall does not allow going back when requirements change. Spiral is heavier and mainly for high-risk projects. Agile could also work, but Prototyping fits best because the main need is to clarify requirements through feedback.

Q 13. Hospital Management System

Question: A hospital plans to develop a Hospital Management System to manage patient registration, appointments, billing, laboratory reports, pharmacy and electronic health records. The hospital requires a secure and reliable system with clearly defined requirements. Major requirement changes are not expected. Identify a suitable software process model and draw its diagram.

Step 1 - Underline these keywords in the case study

clearly defined requirements, major requirement changes are not expected, secure and reliable.

Step 2 - Write the selected model

Selected Software Process Model: Waterfall Model (Classical Life Cycle Model)

Step 3 - Justification (write 3-4 points)

- Requirements are clear and stable, so planning everything first works well.
- Every phase is documented and reviewed before the next one starts, which gives traceability. This matters for patient records and billing.
- Complete testing after implementation helps ensure reliability and security of the system.
- It is simple to understand and manage, with clear milestones and deliverables.

Step 4 - Phases / working of the model

Phase	Activity (with respect to this case study)
1. Requirements	Study patient registration, appointments, billing, lab, pharmacy, EHR. Prepare the SRS.
2. System design	Architecture, database schema, modules, security design (HLD and LLD).
3. Implementation	Code each module and do unit testing.
4. Testing	Integration, system, security and acceptance testing.
5. Deployment	Install the system in the hospital and train staff.
6. Maintenance	Fix bugs, apply updates, adapt to rule changes.

Step 5 - Draw the diagram (neat, labelled, with arrows)

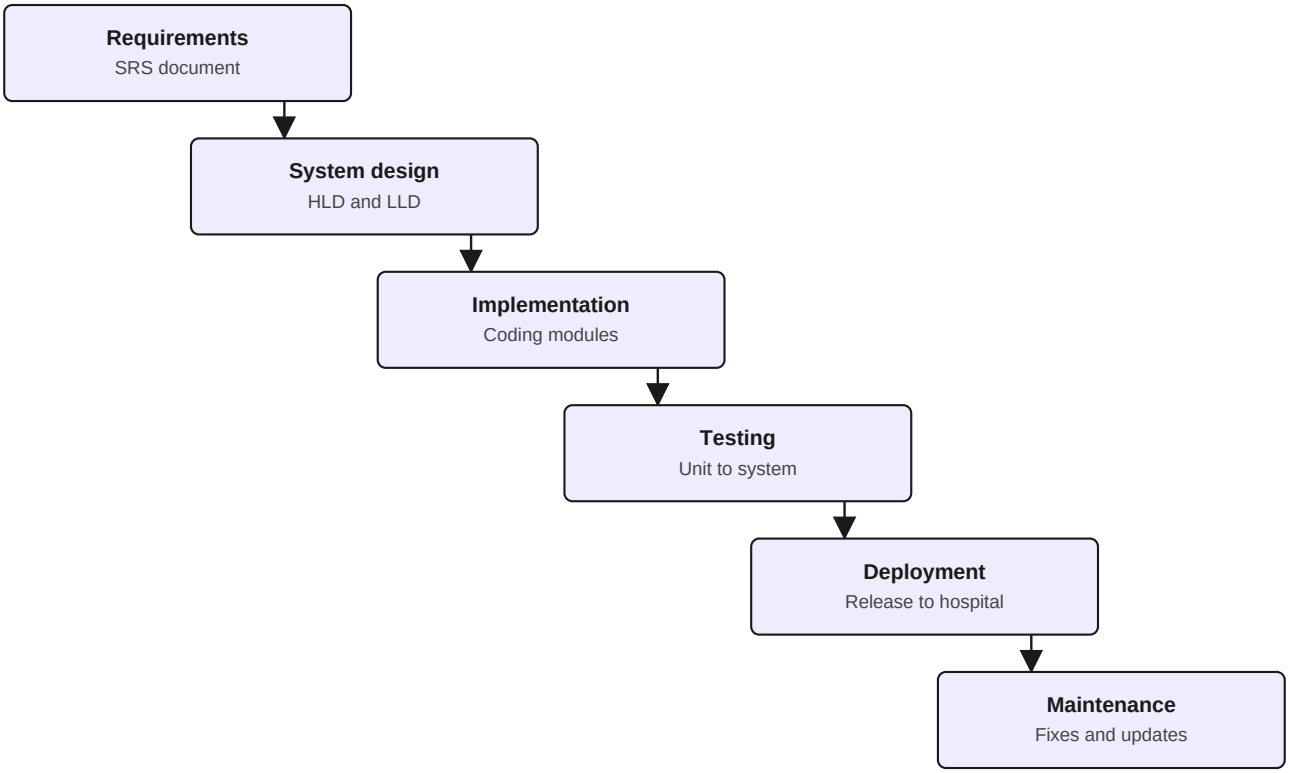

Fig: Waterfall Model

Step 6 - Advantages and disadvantages (2-3 each)

Advantages	Disadvantages
- Simple and easy to manage - Well documented, clear milestones - Suits stable requirements	- No working software until late - Changes are costly after a phase ends - Testing happens late

Step 7 - Conclusion (last line of the answer)

Waterfall Model is the most suitable for the Hospital Management System, because the requirements are clearly defined and stable and the project needs strict documentation, security and reliability.

Viva ready: why not other models?

Agile and Prototyping are meant for changing requirements, which is not the case here. Spiral adds risk-driven cost that is not needed when requirements are stable.

Q 14. Mobile Banking Application

Question: A bank plans to develop a Mobile Banking Application with fund transfer, balance enquiry, bill payment, investment services and biometric authentication. Since financial transactions are involved, the system requires high security, extensive testing and risk management. Identify a suitable software process model and draw its diagram.

Step 1 - Underline these keywords in the case study

financial transactions, high security, extensive testing, risk management, biometric authentication.

Step 2 - Write the selected model

Selected Software Process Model: Spiral Model

Step 3 - Justification (write 3-4 points)

- The Spiral Model does risk analysis in every loop. Fraud, authentication weakness, data leaks and transaction failure can be found and reduced early.
- The app is built in repeated loops. Biometric login comes first, then balance enquiry and fund transfer, then bill payment and investments, and each loop is tested.
- A prototype in each loop reduces uncertainty, especially for biometric integration.
- The bank evaluates every loop, so it suits a large, costly and high-risk project.

Step 4 - Phases / working of the model

Phase	Activity (with respect to this case study)
1. Planning	Set objectives, requirements and constraints for the current loop.
2. Risk analysis	Identify security and technical risks, build a prototype, plan how to reduce them.
3. Engineering	Develop and test that part of the application.
4. Evaluation	The bank reviews the result and plans the next loop.
Loops	Loop 1: concept and authentication. Loop 2: balance enquiry and fund transfer. Loop 3: bill payment. Loop 4: investments. Then final release.

Step 5 - Draw the diagram (neat, labelled, with arrows)

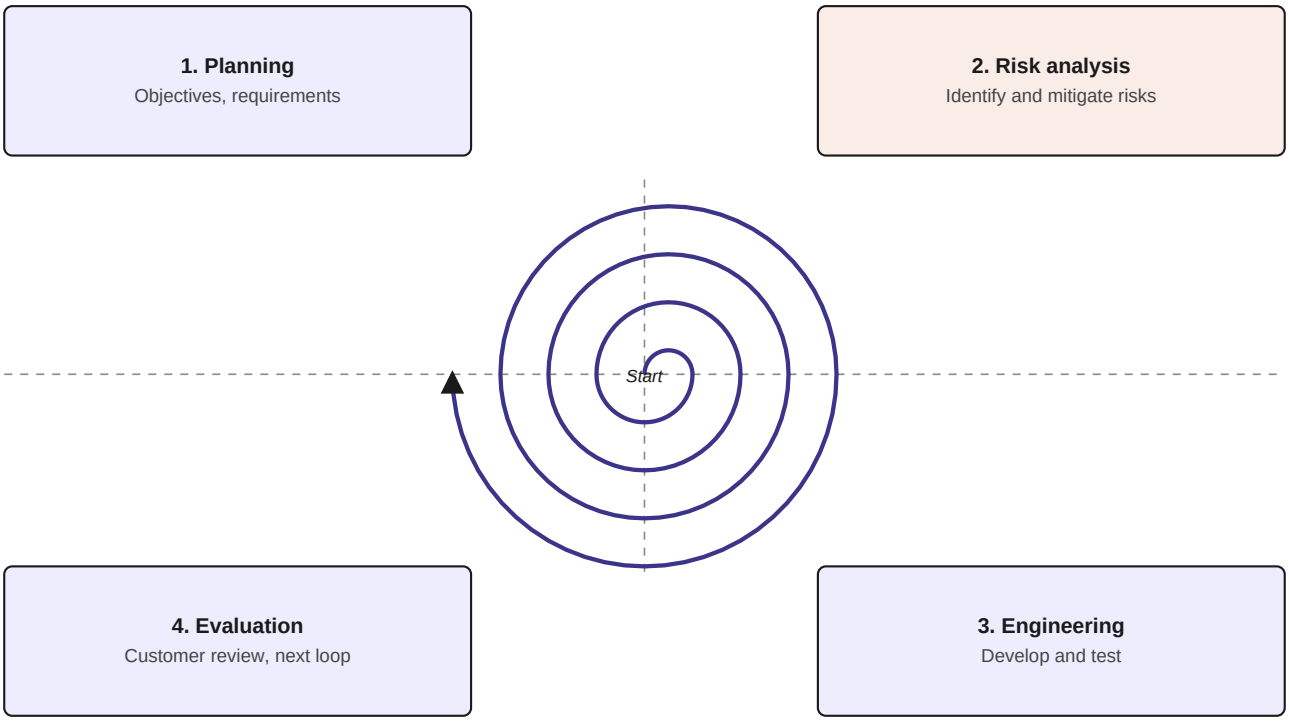

Fig: Spiral Model (clockwise, grows with every loop)

Step 6 - Advantages and disadvantages (2-3 each)

Advantages	Disadvantages
- Strong risk handling - Early and repeated testing - Flexible and suits large projects	- Costly and time consuming - Needs risk analysis experts - Complex to manage, not for small projects

Step 7 - Conclusion (last line of the answer)

Spiral Model is the most suitable for the Mobile Banking Application, because it combines iterative development with risk analysis in every loop, which is essential for a secure financial system.

Viva ready: why not other models?

Waterfall tests only at the end and does not manage risk. Agile is fast but gives less formal risk analysis and documentation, which a bank needs. Prototyping alone does not cover risk and security management.

Q 15. AI-Based Smart Traffic Management System

Question: A city corporation plans to develop an AI-based Smart Traffic Management System to monitor traffic, optimize signal timings, detect accidents and provide real-time route suggestions. The project involves AI integration, IoT devices and evolving requirements. Identify a suitable software process model and draw its diagram.

Step 1 - Underline these keywords in the case study

AI integration, IoT devices, evolving requirements, real-time.

Step 2 - Write the selected model

Selected Software Process Model: Agile Model (Scrum framework)

Step 3 - Justification (write 3-4 points)

- Requirements keep evolving, and Agile accepts changes in every sprint.
- AI models need experiments and tuning on real traffic data, which fits short iterations with feedback.
- Features are delivered step by step: signal optimisation first, then accident detection, then route suggestions.
- The city corporation reviews a working increment after each sprint, and IoT integration problems are found early.

Step 4 - Phases / working of the model

Phase	Activity (with respect to this case study)
Product backlog	Prioritised list of features (monitoring, signal timing, accident detection, routes).
Sprint planning	Team selects items for a 2-4 week sprint and sets a sprint goal.
Sprint	Build and test the items, with a short daily scrum meeting.
Increment	A working, tested part of the system at the end of the sprint.
Sprint review	Demo to the city corporation and collect feedback.
Retrospective	Team discusses what to improve. Backlog is updated for the next sprint.

Step 5 - Draw the diagram (neat, labelled, with arrows)

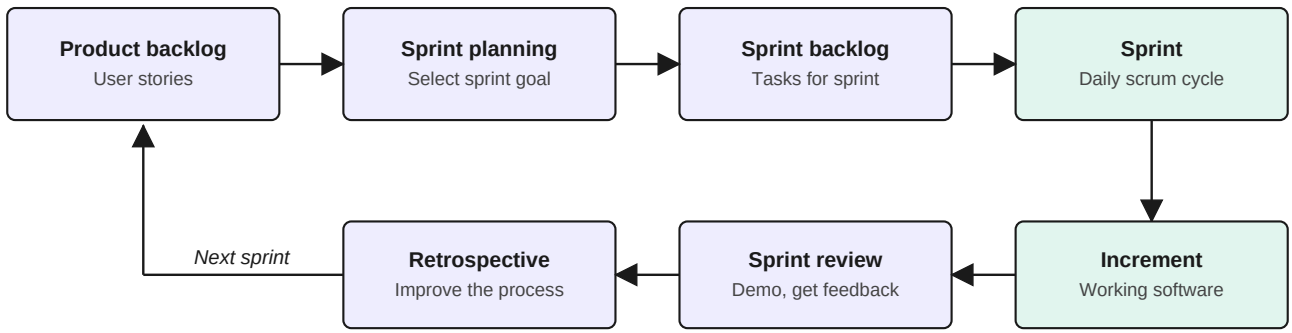

Fig: Agile (Scrum) Model

Step 6 - Advantages and disadvantages (2-3 each)

Advantages	Disadvantages
- Adapts easily to change - Working software delivered early - Strong customer involvement	- Less documentation - Cost and time hard to predict - Needs an experienced, well-communicating team

Step 7 - Conclusion (last line of the answer)

Agile (Scrum) is the most suitable for the Smart Traffic Management System, because the requirements evolve and the AI and IoT parts need frequent feedback and incremental delivery.

Viva ready: why not other models?

Waterfall cannot handle changing requirements. Spiral is heavier on risk documentation and cost. Prototyping builds a sample, but this project needs a continuously delivered, working product.

Roles in Scrum: Product Owner, Scrum Master, Development Team. Artifacts: Product Backlog, Sprint Backlog, Increment.